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NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY 9744/02

Paper 2 Structured Questions **September 2017**

Candidates answer on the Question Paper.

No Additional Materials are required. **2 hours**

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do no use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do no use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
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2	
3	
4	
5	
6	
7	
8	
9	
10	
Total	

This document consists of **21** printed pages and **1** blank page.

[Turn over

Answer **all** the questions in this section.

1 **Table 1** shows some features of four biological molecules that are all polymers.

(a) Complete Table 1 by using a tick (✓) to indicate the features that apply to each polymer.

Table 1

feature	amylopectin	cellulose	RNA	polypeptide
synthesised from amino acid monomers				
contains glycosidic bonds				
polymer is branched				
contains nitrogen				
can be found in both animal and plant cells				

[4]

(b) **Fig. 1** is a simple diagram of a phospholipid molecule.

Explain how the structure of a phospholipid molecule makes it suitable for its function in cell membranes. You may label and annotate **Fig. 1** as part of your answer.

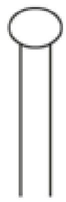

Fig. 1

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[3]

(c) State two components of a cell surface membrane other than phospholipid molecules and describe their function.

component 1 ....

function ....

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component 2 ....

function ....

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[4]

The fluid mosaic model of membrane structure was first proposed in 1972 by Singer and Nicolson. The model describes in detail how the components of a membrane are organised.

(d) Suggest why 'fluid mosaic' is an appropriate term to use to describe membrane structure.

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[3]

[Total: 14]

2 Erythropoietin, also known as EPO, is a large glycoprotein synthesised by specialised cells in the kidney. These cells are very sensitive to changes in oxygen concentration in the blood passing through the kidney and respond to a low oxygen concentration by increasing the synthesis of EPO.

EPO acts at the surface of particular target cells, such as cells in the bone marrow. These bone marrow cells are stimulated to produce red blood cells.

(a) A low oxygen concentration leads to an increase in the quantity of mRNA in the specialised cells in the kidney by activating hypoxia-inducing factors (HIFs) which regulates gene expression at the transcriptional level.

Explain how the activation of HIFs results in an increase in EPO mRNA in the cells.

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[3]

(b) All cells of the body are exposed to circulating blood plasma containing EPO, but only particular target cells respond.

(i) Suggest why EPO acts on target cells and not other cells.

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[2]

(ii) EPO cannot pass through the cell surface membrane to enter the bone marrow cells. Suggest **one** reason why this is so.

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[1]

(c) Red blood cells originate from undifferentiated cells in the bone marrow that are capable of continuous mitotic cell division.
State the name of this type of undifferentiated cell.

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[1]

[Total: 7]

3 (a) **Fig. 3** is a diagram of an ATP molecule.

Label **Fig. 3** to show the components of an ATP molecule.

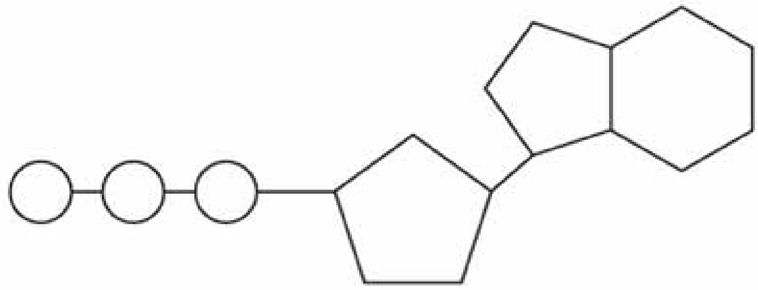

Fig. 3 [3]

(b) ATP is used during translation in amino acid activation, when an amino acid becomes attached to its specific tRNA molecule having a particular anticodon. The reaction requires an enzyme called aminoacyl tRNA synthetase.

(i) Explain why a particular amino acid needs to be linked to a specific tRNA molecule.

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.... [2]

(ii) Explain how the structure of an enzyme such as aminoacyl tRNA synthetase would be altered if the pH of the cytoplasm became too acidic.

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.... [2]

(iii) Aminoacyl tRNA synthetase uses the induced fit mechanism.

Explain the induced fit mechanism.

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.... [2]

[Total: 9]

4 *Morbillivirus* causes measles. The structure of *Morbillivirus* is shown in **Fig. 4**.

Haemagglutinin (H) and fusion protein (F) are glycoproteins embedded in the viral envelope.

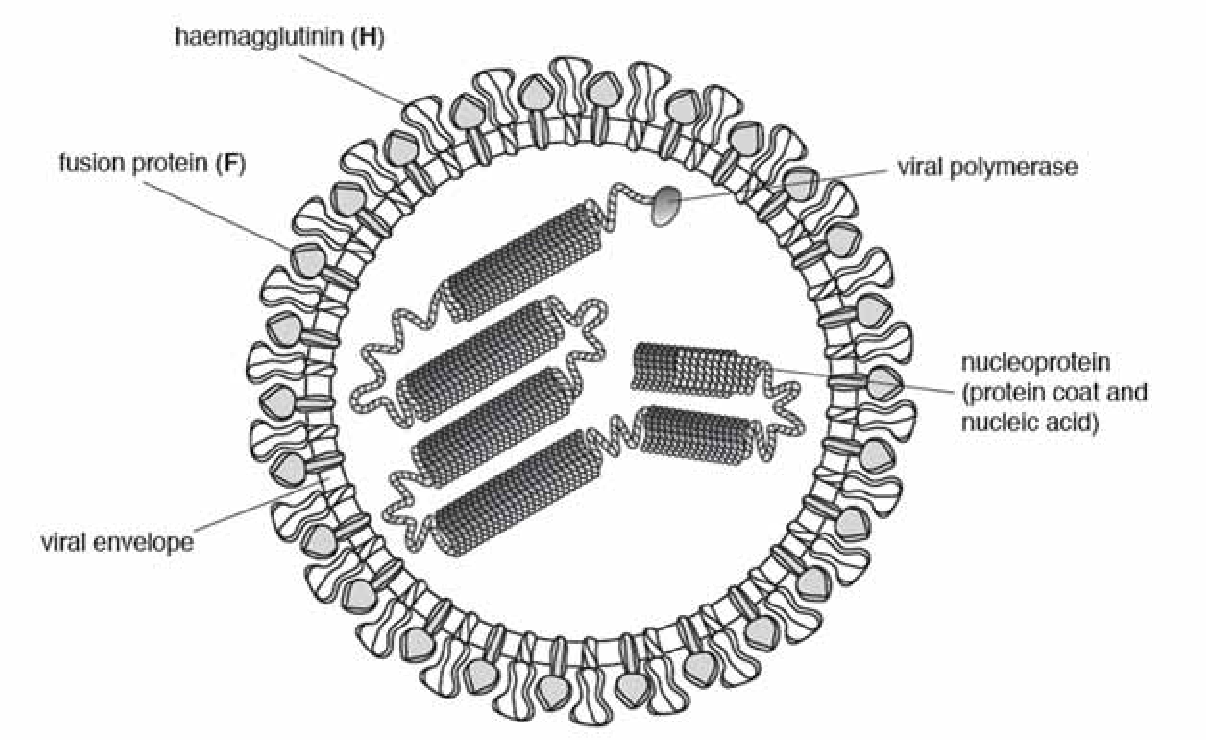

Fig. 4

Morbillivirus only infects cells that have a membrane glycoprotein known as signalling lymphocyte activation molecule (SLAM).

When *Morbillivirus* infects a cell, **H** acts before **F**. After the virus binds to the host cell, only the nucleoprotein with the viral polymerase enters the host cell and the virus is replicated.

Morbillivirus replicates its genetic material in the same manner as influenza virus.

New viral particles leave the host cell by budding from the cell surface membrane of the cell. This forms the main part of their envelope.

(a) List two ways in which the structure of *Morbillivirus*:

(i) is similar to HIV

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.... [2]

(ii) differs from the HIV

[2]

(b) With reference to **Fig. 4.1** and the information provided,

(i) suggest how *Morbillivirus* infects a cell with SLAM glycoproteins so that only nucleoprotein and viral polymerase enter.

[2]

(ii) suggest the role of viral polymerase in *Morbillivirus*.

[3]

(c) Measles has only one serotype. Within this serotype, there are 24 genotypes recognised to date.

Describe how these 24 genotypes could have arisen.

[2]

[Total: 11]

5 An *Allium cepa* root was cut into ten transverse sections at different distances from the tip. The sections were stained and viewed under the microscope. The number of cells in mitosis were counted in each section and the results were used to determine the mitotic index.

This is calculated as follows:

$$\text{mitotic index} = \frac{\text{number of cells in mitosis}}{\text{total number of cells}}$$

Fig. 5.1 shows the mitotic index for the ten sections.

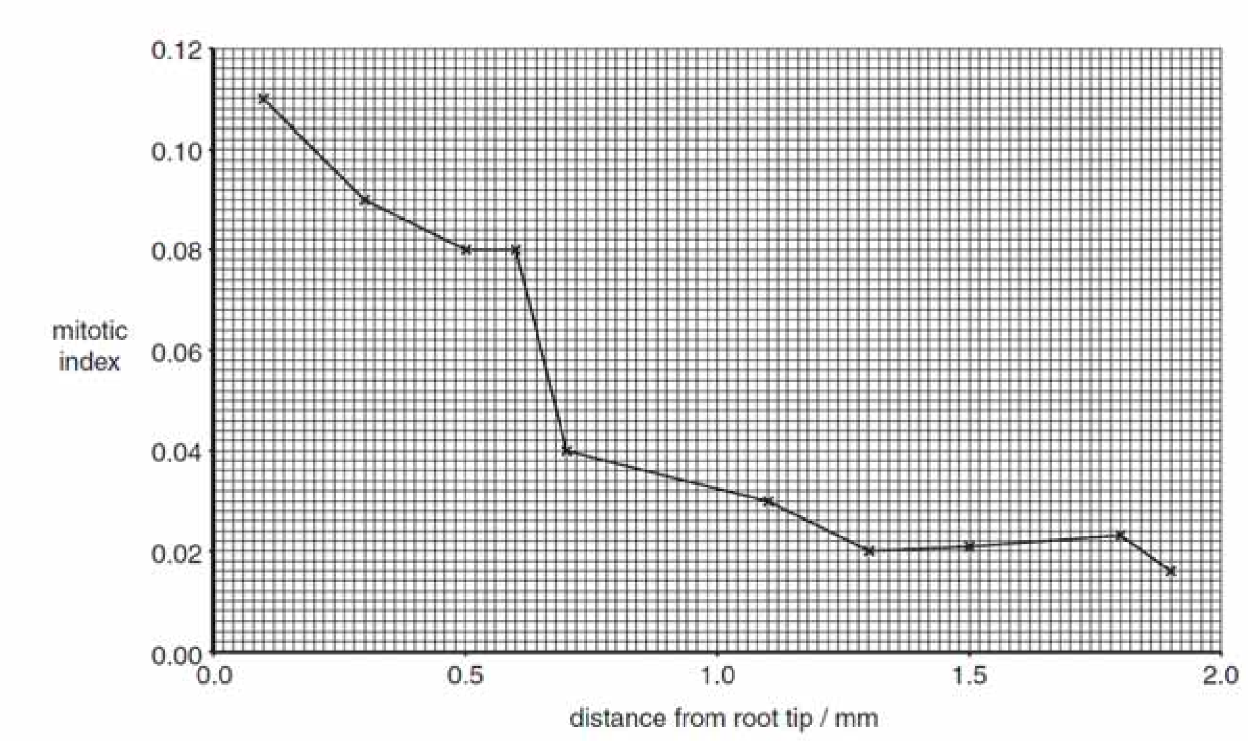

Fig. 5.1

(a) Using the information in **Fig. 5.1**, describe how the mitotic index changes along the length of the root.

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.... [2]

(b) Explain how the events in the mitotic cell cycle ensure that all the cells in the root are genetically identical.

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.... [3]

(c) Two genes, **A/a** and **B/b** are linked on the same pair of homologous chromosomes in *Allium cepa*, as shown in **Fig. 5.2**.

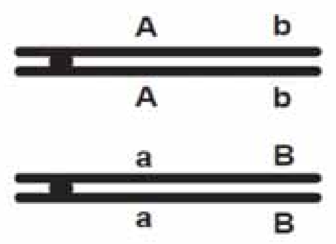

Fig. 5.2

With reference to **Fig. 5.2**,

(i) draw a diagram to show the effect of crossing-over between the homologous chromosomes.

[2]

(ii) state the effect of linkage and crossing-over on the proportions of gametes with different genotypes that are produced.

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.... [2]

[Total: 9]

6 *CFTR*, the cystic fibrosis (CF) gene, encodes for the CFTR protein that plays an essential role in anion regulation and tissue homeostasis of various epithelia. In the gastrointestinal tract CFTR promotes chloride and bicarbonate secretion, playing an essential role in ion and acid-base homeostasis.

Gene mutations responsible for cystic fibrosis can be classified into 6 different classes.

- **Class I** mutations can result from nonsense and frame-shift mutations, as well as mRNA splicing defects.
- **Class III and IV** mutations are typified by aberrant channel function, rather than reduced quantities of CFTR.

(a) Explain how **Class I** mutations result in a reduction in the quantity of functional CFTR protein.

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[2]

(b) Explain how **Class III** and **IV** mutations result in aberrant channel function.

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[2]

CFTR, has been identified as a candidate driver gene for colorectal cancer in mice and humans. Further, recent epidemiological and clinical studies indicate that CF patients are at high risk for developing tumours in the colon. Investigations suggest that *CFTR* is a ***tumour suppressor gene*** in the intestinal tract as *CFTR* mutant mice developed significantly more tumours in the colon and the entire small intestine.

(c) Explain why a mutation in the tumour suppressor gene, such as the *CFTR* gene, may lead to significantly more tumours in mice.

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.... [3]

(d) Suggest how tumours can eventually become malignant, leading to cancer.

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.... [2]

[Total: 9]

7 **(a)** Sometimes a gene has more than two alleles, termed *multiple alleles*.

The ABO blood group system in humans is controlled by a gene with three alleles, I^A, I^B and I^o. Alleles I^A and I^B are codominant and I^o is recessive to both.

The blood group **AB** is the result of codominance.

Explain what is meant by *codominance* in this context.

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[3]

(b) In humans, a gene that codes for the production of a protein, called factor VIII, is located on the X chromosome. The dominant allele for this gene produces factor VIII, but the recessive allele does not produce factor VIII.

A person who is unable to make factor VIII has haemophilia in which the blood fails to clot properly.

Explain why a man with haemophilia cannot pass haemophilia to his son but may pass haemophilia to his grandson.

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[3]

(c) A gene for feather colour in chickens is carried on an autosome. This gene has two alleles, black (**C**$^{\mathbf{B}}$) and splashed-white (**C**$^{\mathbf{W}}$). When a male chicken with black feathers is mated with a female chicken with splashed-white feathers, all the offspring have blue feathers. This also occurs when a male chicken with splashed-white feathers is crossed with a female with black feathers.

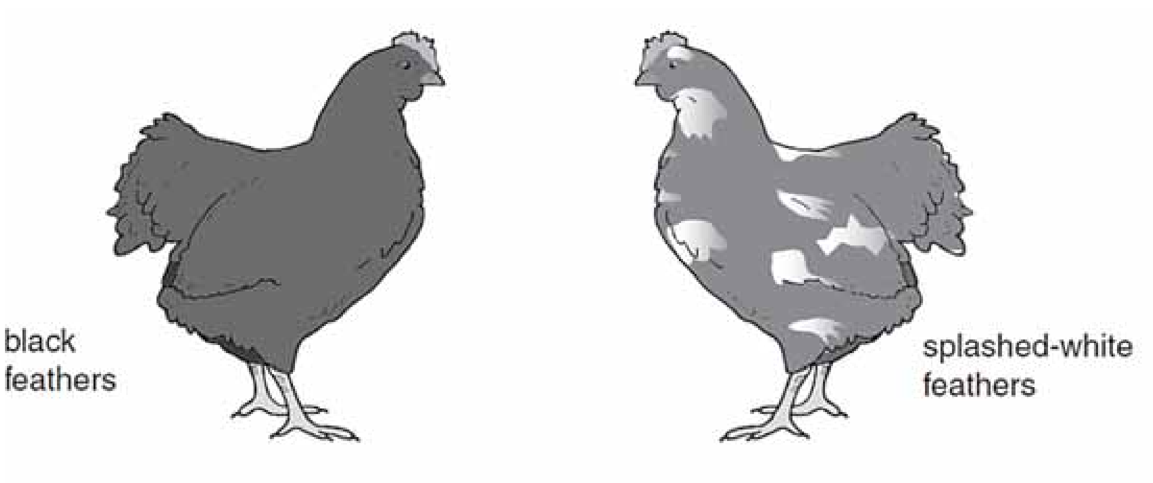

Fig. 7.1

Another gene may cause stripes on feathers (barred feathers). This gene is carried on the X chromosome. The allele for barred feathers (**X**$^{\mathbf{A}}$) is dominant to the allele for nonbarred feathers (**X**$^{\mathbf{a}}$).

In chickens, the male is homogametic and has two X chromosomes while the female is heterogametic and has one X chromosome and one Y chromosome.

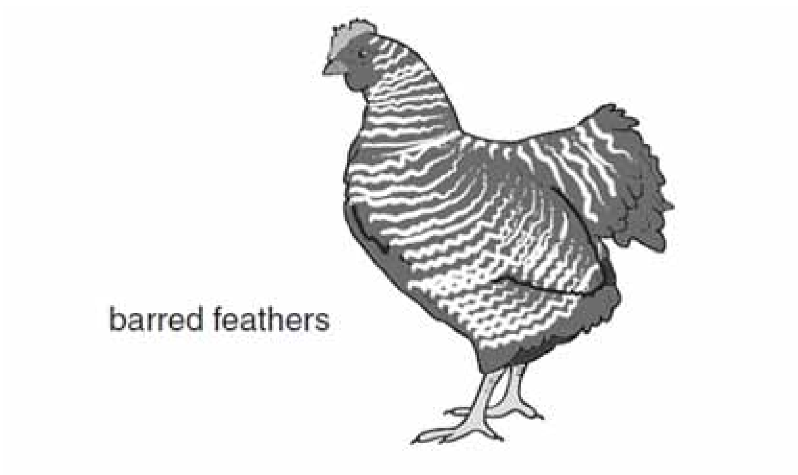

Fig. 7.2

(i) A male chicken with black, non-barred feathers was crossed with a female chicken with splashed-white, barred feathers. All the offspring had blue feathers, but the males were barred and the females were non-barred.

Using the symbols given above, draw a genetic diagram to show this cross.

parents' phenotype	male, black, non-barred feathers.	female, splashed-white, barred feathers.
genotype		
gametes		
offspring genotypes		
phenotypes	male, blue, barred feathers.	female, blue, non-barred feathers.

[3]

(ii) Explain how a farmer could use a breeding programme to find out the genotype of a male chicken with blue, barred feathers.

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[3]

[Total: 12]

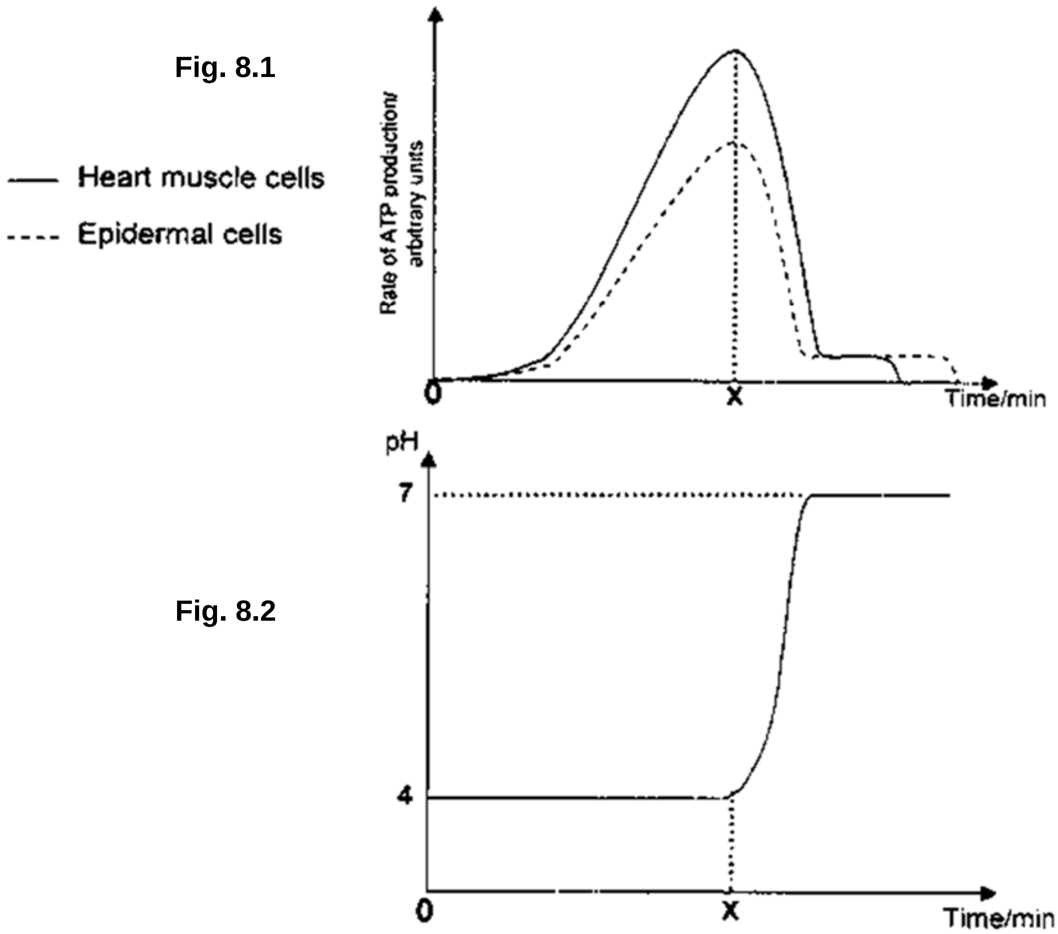
8 Heart muscle cells and epidermal cells were extracted from Chinese hamsters. The cells were lysed and the mitochondria and cytosol were isolated. The mitochondria and cytosol were then mixed and re-suspended in a culture of essential nutrients. This suspension system was used to study the process of cellular respiration.

At time **0**, glucose was added to the system. At Time **X**, digitonin, a detergent which disrupts membranes was introduced to the suspension system. A probe was used to measure the concentrations of ATP as well as the pH level in the mitochondria.

The experimental results are recorded in the graphs shown.
Fig. 8.1 shows the rate of ATP production for heart muscle cells and epidermal cells.
Fig. 8.2 shows the pH level of the mitochondria in both heart muscle and epidermal cells.

Fig. 8.1

Fig. 8.2



(a) Account for the difference in the level of ATP production in both tissues after glucose was added.

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.... [1]

(b) With reference to **Fig. 8.1**, explain the changes in ATP production over time for the heart muscle cell suspension.

[3]

(c) With reference to **Fig. 8.2**, state which region of the mitochondrion the pH probe was measuring. Explain your conclusion.

[2]

(d) Suggest why cytosol was used to re-suspend the mitochondria.

[1]

(e) From your biological knowledge, explain the adaptation of the double membrane for its role in the production of energy.

[2]

[Total: 9]

9 The camel family, Camelidae, are well-known for their ability to survive the hot and dry conditions of the desert, but studies have found that they once thrived in colder climates. A recent finding by a group of scientists unearthed fossilised remains of a giant species of camel in North America. This giant species of camel is closely-related to the one-humped Dromedary camel now found in Africa (**Fig. 9.1**). The distinctive humps of both camels are fat stores.

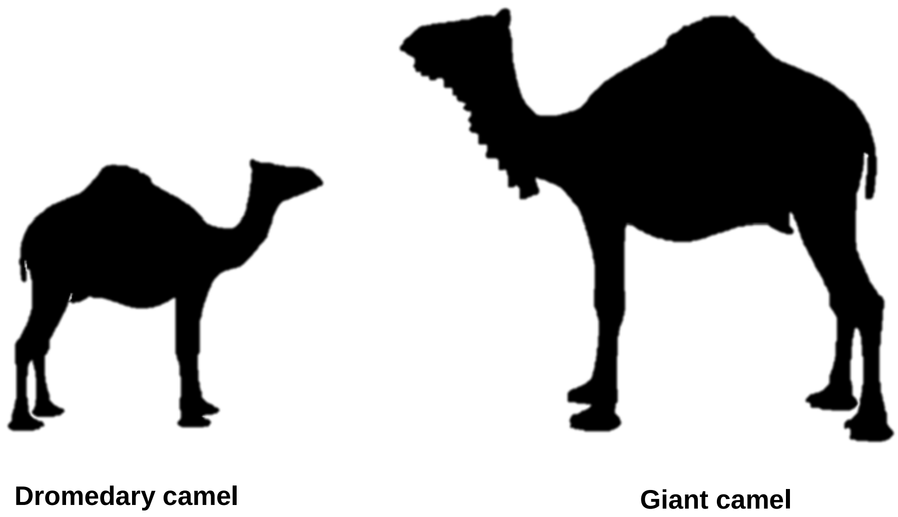

Fig. 9.1

(a) Suggest how the presence of hump in both the ancestral Giant camel and modern-day Dromedary camel allow them to adapt to their respective habitats.

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.... [2]

Until only about two or three million years ago, the Camelids were largely confined within North America. Following the formation of the Bering Land Bridge and the Isthmus of Panama, camels migrated from North America to Asia and South America respectively. Today, camels are no longer found in North America.

Three modern-day groups of species survive today:

- the one-hump Dromedary camel of north Africa and southwest Asia,
- the two-hump Bactrian camel of central Asia and
- the South American Camelids group which has diverged into four species: llamas, alpacas, guanacos, and vicuñas.

Fig. 9.2 shows the distribution of modern-day camels in the world.

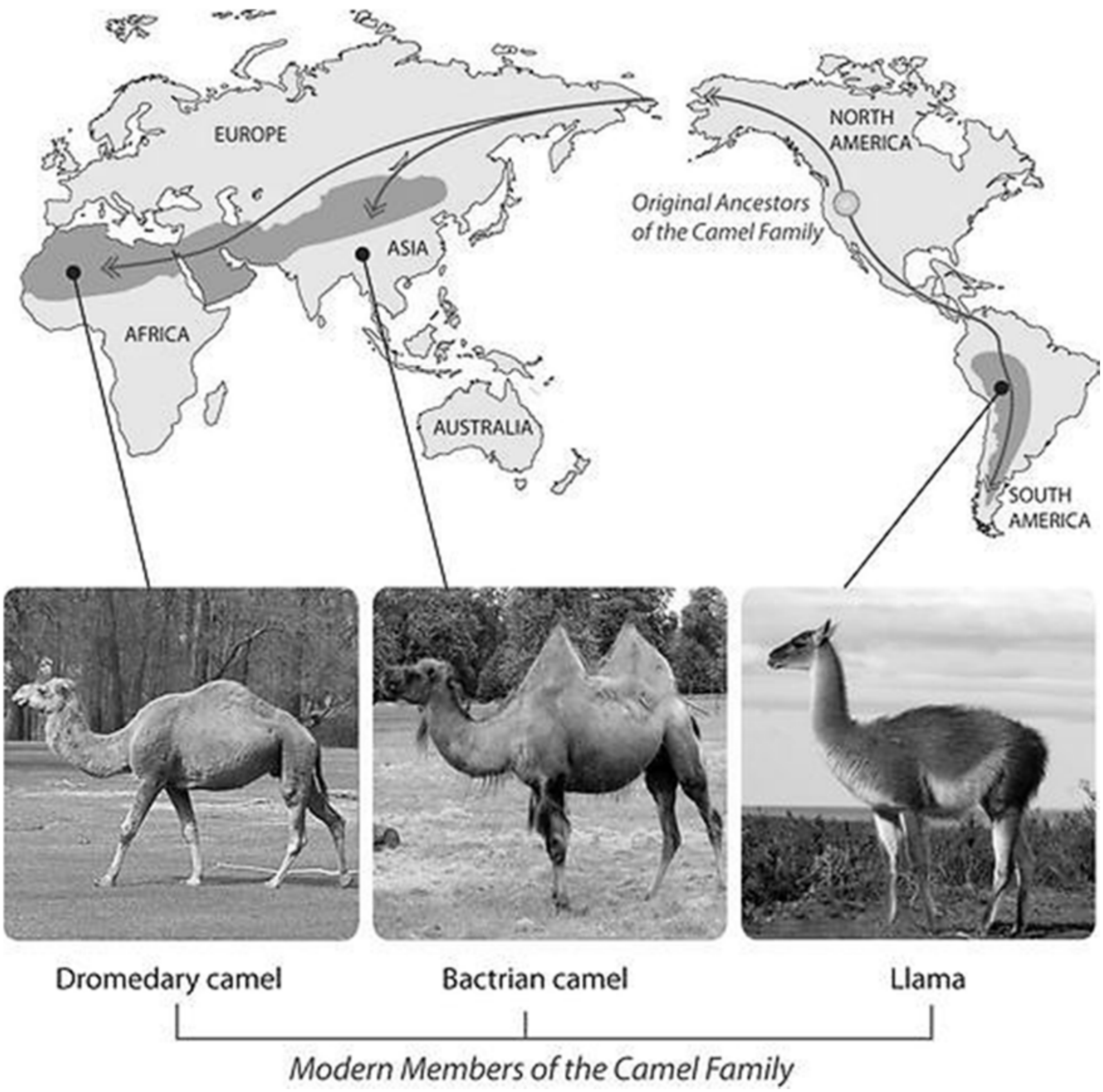

Fig. 9.2

The various species of the modern-day camels evolved from the ancestral population in North America.

(b) With reference to **Fig. 9.2**, describe how natural selection could have occurred to give rise to the various species of camels in the world today.

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.... [4]

(c) The Arabian camels in Australia typically have sand coat colour. However, there is a small percentage of camels which have albino coat. It is known that the albino coat is a recessive condition and it reduces the life-span of the camel. The population of the sand coat and albino coat camels in a region was documented.

(i) Explain why population is considered the smallest unit of evolution.

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.... [2]

(ii) Explain whether it is likely that the albino camel will disappear completely from the population over time. Assume there is no introduction of new albino camels into the population.

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.... [2]

[Total: 10]

10 (a) The infectious disease tuberculosis (TB) is caused by a bacterium.

Each of the descriptions **A** to **C** describes a cell structure found in prokaryotic cells **and** in plant cells.

For each of the descriptions **A** to **C**:

- name the cell structure described
- state **one** difference in this structure between a prokaryotic cell and a plant cell.

A the site of polypeptide synthesis

cell structure ....

difference ....

....

B the genetic material of the cell

cell structure ....

difference ....

....

C the structure that provides a rigid shape to the cell and prevents osmotic lysis

cell structure ....

difference ....

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[3]

(b) Describe how TB is transmitted from an infected person to an uninfected person.

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[2]

(c) Outline how the use of vaccine can give protection against diseases such as tuberculosis.

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[5]

[Total: 10]